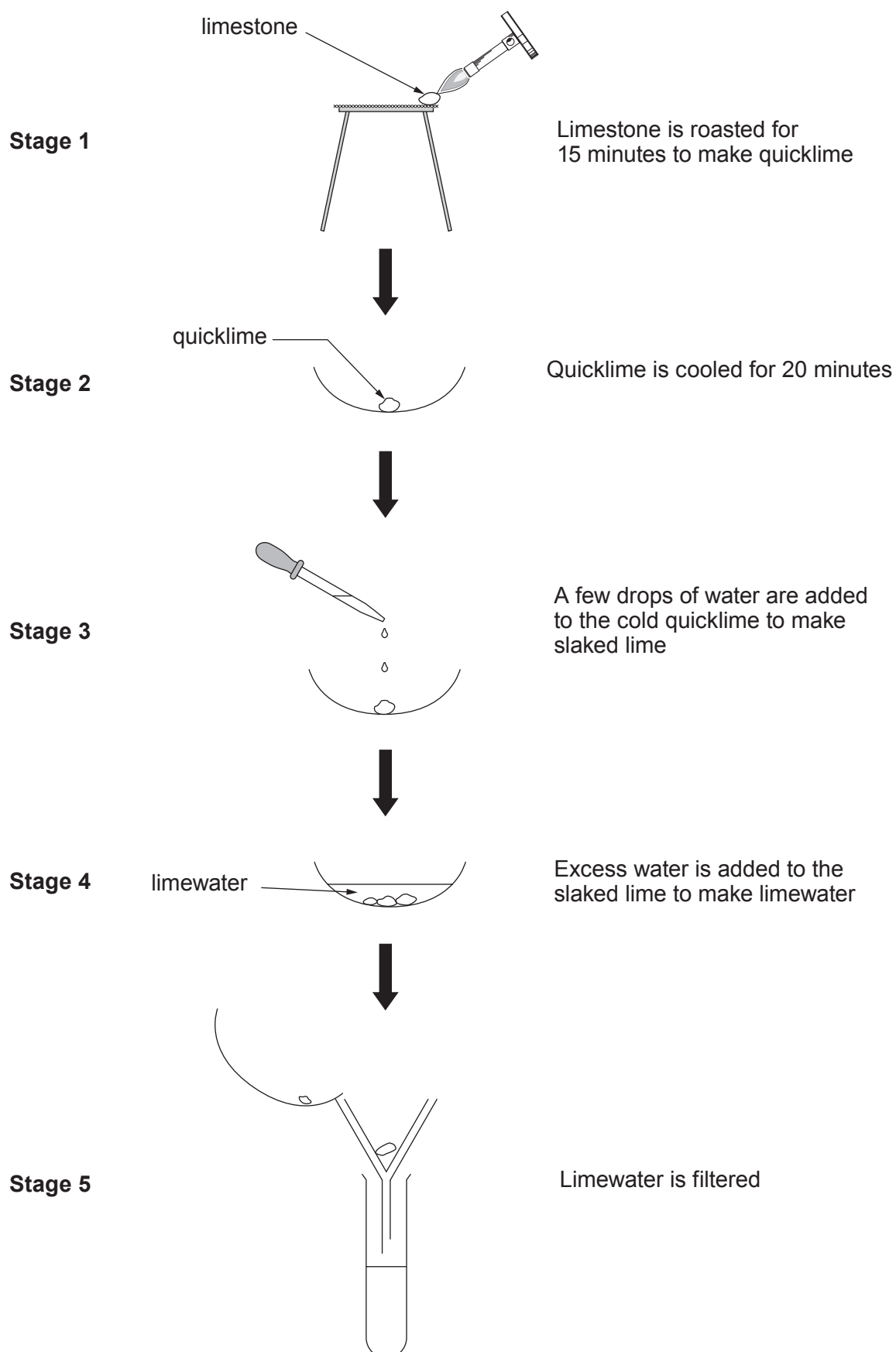


7. The flow chart shows the stages a student carried out to change calcium carbonate (limestone) into calcium hydroxide solution (limewater).



(a) Use the information in the flow chart to answer parts (i) to (iii).

(i) Name the type of reaction taking place in stage 1.

[1]

.....

(ii) Give the **number** of the stage which shows an exothermic reaction.

[1]

.....

(iii) I. Describe what you would expect to **see** happen in stage 3.

[2]

.....

.....

.....

II. The reaction taking place in stage 3 is shown by the following word equation.

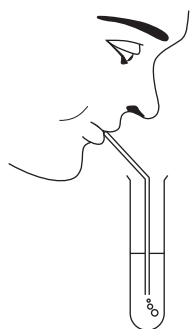


Write a balanced **symbol** equation for this reaction.

[2]

..... + $\longrightarrow$

(b) The student blew gently through a straw placed in a test tube containing limewater.



Describe what you would expect to happen to the limewater. Give a reason for your answer.

[2]

Observation

Reason

- (c) Quicklime is manufactured from limestone in a lime kiln. A manufacturer expected to obtain 5.6 tonnes per 10 tonnes of limestone used. The actual mass of quicklime produced was 5.1 tonnes.

Calculate the percentage yield of quicklime for this manufacturing process. [2]

percentage yield = %

Examiner
only

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